EDITORS
NORITA AHMAD American University of Sharjah; nahmad@aus.edu
PREETI CHAUHAN IEEE Reliability Society; preeti.chauhan@ieee.org

Redefining Human Resource Practices With AI Agents and Agentic AI: Automated Compliance and Enhanced Productivity

Nir Kshetri, University of North Carolina at Greensboro

This article analyzes how agentic artificial intelligence is revolutionizing human resource management through automated workflows, enhanced decision making, and improved employee experiences while addressing implementation challenges like security risks, regulatory compliance, and workforce adoption.

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Agentic artificial intelligence (AI) is revolutionizing human resource management (HRM) by automating routine tasks such as recruitment, thereby streamlining workflows, minimizing bias, and enhancing overall efficiency.[1] Additionally, it uses predictive analytics to assess turnover risks, enabling proactive intervention and cost savings. This technology is also likely to boost employee satisfaction and improves retention strategies.[1] Agentic AI thus enables strategic HRM with tailored, goal-oriented services.[2]

Agentic AI allows users to define a task, which the AI autonomously breaks into steps, designs workflows, and selects appropriate AI or IT services for execution.[3] As an illustrative case, we can consider a task to improve employee retention. An AI HR agent autonomously streamlines HR processes by identifying improvement strategies, generating task lists, and collecting relevant data, from analyzing exit interviews

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to benchmarking industry best practices. It systematically stores insights in a centralized knowledge base, refining its approach through iterative feedback loops. For instance, after conducting employee satisfaction surveys, the agent cross-references findings with turnover metrics, updates retention strategies, and adapts future actions based on outcomes. This closed-loop automation enhances both operational efficiency and evidence-based decision-making in HR.[4]

Agentic AI solutions benefit not only large enterprises but also small and medium-sized enterprises (SMEs), enhancing operational efficiency and competitiveness across the board. SMEs can enhance their HR operations by implementing agentic AI, which automates and refines various HR functions. In talent acquisition, AI agents can autonomously screen resumes, evaluate candidate qualifications, and generate shortlists of top applicants, thereby expediting the recruitment process and minimizing manual effort. Additionally, agentic AI can streamline onboarding by guiding new hires through necessary documentation and training modules, ensuring a smooth integration into the company.[5]

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Agentic AI not only automates tasks to boost employee productivity but also offers job seekers personalized career guidance, enhancing their empowerment beyond what generative AI provides. Generative AI assists by generating a cover letter based on a provided résumé and job description. In contrast, agentic AI autonomously creates a LinkedIn account, drafts applications, applies to all suitable jobs, and provides a secured record of its actions.[6]

This article explores how agentic AI is revolutionizing HR, automating workflows, enhancing decision making, and improving employee experiences, while addressing key challenges like security, bias, and adoption resistance. It highlights real-world implementations, regulatory impacts, and future-ready strategies for ethical AI integration in HRM.

AGENTIC AI SOLUTIONS FROM LEADING HR TECHNOLOGY COMPANIES

HR technology vendors are increasingly integrating agentic AI into their solutions to enhance various human resource functions and streamline operations (Table 1). This trend signifies a major technological shift in HR practices. These innovations include AI agents that coordinate tasks across various systems and departments, manage AI agents throughout the enterprise, and automate routine HR tasks, allowing professionals to focus on more strategic work.

TABLE 1. Agentic AI and AI agents developed by leading HR technology companies and their applications.

Company	Agentic AI/AI agents	HRM function	Functionality
ServiceNow	In AI Agent Studio, the AI Agent Orchestrator serves as the central coordinator, managing workflows across multiple AI agents and enterprise systems.[27]	Onboarding	Automatically delivers pertinent information to new employees and addresses their inquiries, streamlining the onboarding process.[28]
Workday	Recruiting Agent	Recruitment	Enhances talent acquisition by automating tasks such as job description creation, candidate sourcing, and interview scheduling; integrates with platforms like Microsoft Teams for streamlined communication.[29]
UKG	Bryte AI Agents	Employee training and development	Proactively solve problems and make recommendations through multistep processes that learn and improve over time, assisting in managing promotions and identifying impacts of new tax regulations.[7]
Oracle	Career Planning Guide, Performance and Goals Assistant	Career development, performance management, learning and development	Recommends career paths tailored to employees' skills and experience; monitors goal progression, offering timely suggestions to ensure employees remain aligned with their performance objectives ahead of evaluations.[30]

ServiceNow has launched the AI Agent Orchestrator to coordinate specialized AI agents across tasks, systems, and departments, along with thousands of prebuilt agents for HR, plus the AI Agent Studio for creating custom agents.[7] The recruitment sector is at the forefront of leveraging advanced AI agents.[8] Workday offers AI-powered agents to improve HR processes, including recruiting and succession planning, and has introduced the Agent System of Record to manage these agents effectively.[7] Oracle has developed AI agents within its Cloud HCM to streamline HR processes, assisting employees with career development, time-off requests, and workforce analytics, while providing HR teams with centralized data insights.[9]

Traditional HR systems, primarily databases for payroll and compliance, are thus evolving into integrated talent intelligence systems through the incorporation of intelligent agents into core human capital management platforms.[10] Beyond the HR technology providers highlighted in Table 1, numerous other enterprises have deployed agentic AI solutions to automate and optimize HRM functions. NEC launched agentic AI to automate talent management decisions. The system conducts comprehensive internal and external data searches to optimize these processes.[3] LinkedIn's 2024 release of the Hiring Assistant marked a shift toward AI-driven automation in talent acquisition.[8]

ENHANCING HR SELF-SERVICE AND EMPLOYEE EXPERIENCES

Employee experience covers all workplace interactions, from onboarding to daily work and advancement, impacting engagement and retention.[11] Fragmented HR systems often lead to employee frustration, but integrating HR self-service can enhance the employee experience by providing accessible information and empowering employees.[12] Modern employees expect user-friendly technologies and tend to reject those they dislike. Demonstrating a dedication to enhancing employee experience, 87% of HR managers aim to raise their HR tech budgets in 2024.[16]

In general, AI-driven systems enable employees to swiftly access information via simple queries, eliminating the need to navigate multiple platforms or submit service requests. This enhances satisfaction and reduces operational costs, streamlining HR processes while improving user experience.[16] Building upon earlier AI, agentic AI automates tasks and streamlines workflows, providing a smoother, more engaging technological experience.[16] AI agents such as Galileo, Microsoft Copilot, Workday Assistant, and Eightfold's AI agent streamline HR operations by automating tasks, improving employee experiences, and reducing integration efforts for HR teams.[13] Integrating AI agents into HR systems will eliminate the need for employees to navigate multiple platforms, significantly enhancing the HR tech experience.[10]

Additionally, AI agents enhance employee experiences by personalizing training and career development recommendations, assigning role-specific modules to new hires to equip them with necessary skills.[1] For instance, ZBrain's Training Module Assignment Agent analyzes job roles and employee data to deliver customized and efficient learning experiences.[4]

Agentic AI also enhances employee experiences by ensuring employees maximize their benefits, such as paid vacation. This personalization fosters a positive employee experience. Traditional HR chatbots answer basic questions like available vacation days. In contrast, AI agents not only provide this information but also guide employees through time-off requests and assist in booking travel arrangements. Seamlessly integrating into HR systems, these agents offer a dynamic and proactive approach to workforce management.[17]

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BOOSTING OPERATIONAL EFFICIENCY AND PRODUCTIVITY

Agentic AI is transforming business operations by automating complex tasks and integrating AI-driven processes across multiple systems, leading to significant efficiency and productivity gains across various industries.[10] Accenture deploys AI-powered scheduling assistants (Agentforce) to optimize employee productivity through automated work planning, priority identification, and office time utilization.[14] Agentic AI tools also automate compliance processes by monitoring policy adherence, flagging violations, and providing real-time guidance, dramatically reducing manual oversight while improving accuracy. For instance, ZBrain's compliance agent autonomously audits financial transactions against corporate policies, detecting noncompliant activity with precision to mitigate risks and operational inefficiencies.[4]

AI-powered chatbots have transformed HR by streamlining employee queries and reducing routine tasks. Now, AI agents promise even greater autonomy and operational efficiency.[17] For instance, Chipotle cut hiring time by 75% using Paradox's AI agent, enabling fully automated

hiring; some new hires even mistake the bot ("Amelia") for a human.[8]

AI agents with generative capabilities enhance HR modules, improving decision making and operational efficiency. This transformation enables HR systems to proactively offer deeper insights and automate functions like talent management and recruitment.[10] AI-powered scheduling tools can boost efficiency, with some leaders predicting they will enable widespread four-day workweeks.[18]

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WORKPLACE SAFETY AND COMPLIANCE

HRM researchers have found that employee outcomes in high-risk workplaces lacking safety measures and compliance are adversely affected, leading to increased injuries, decreased job satisfaction, and higher turnover rates.[15] HRM's integration of safety culture into organizational policies shapes employee values and behaviors toward proactive safety practices.[16]

Despite engineering advancements that have reduced workplace hazards, employee safety remains a significant concern. While the physical environment plays a role, human behavior is a critical factor in most workplace accidents and injuries. Risky actions, failures in detecting hazards, and a lack of proactive safety measures contribute to these incidents. Implementing behavior-based safety programs can address these issues by systematically observing and modifying unsafe behaviors, thereby enhancing overall workplace safety.[17] In 2023, private industry employers reported approximately 2.6 million nonfatal workplace injuries and illnesses. Additionally, there were 5,283 fatal work-related injuries across all sectors.[18]

AI boosts workplace safety by predicting hazards via real-time data analysis, monitoring compliance, and preemptively mitigating risks ("Exhibit 1: From Detection to Action—viAct's Integrated Artificial Intelligence System for Real-Time Construction Risk Management"). In construction, AI agents analyze video feeds to detect unsafe behaviors (for example, missing personal protective equipment) and historical data to forecast dangers, while adaptive learning improves responses over time. For example, safety AI agents use the Internet of Things (IoT) and cameras to automate incident reporting and enforce protocols, ensuring a safer, compliant worksite.[19]

Occupational safety AI agents surpass basic chatbots by operating as adaptive, intelligent systems. Leveraging large language models (LLMs)

EXHIBIT 1: FROM DETECTION TO ACTION—viAct'S INTEGRATED ARTIFICIAL INTELLIGENCE SYSTEM FOR REAL-TIME CONSTRUCTION RISK MANAGEMENT

viAct's artificial intelligence (AI)-powered safety system for construction sites implements a three-phase evolution: predictive AI detects hazards through data analysis, generative AI formulates customized risk solutions, and agentic AI (viGent) executes autonomous safety interventions. This integrated approach progressively transforms passive monitoring into proactive protection, enhancing both worker safety and worksite efficiency through intelligent automation. The system's adaptive capabilities enable continuous improvement in hazard prevention and compliance enforcement.[25]

viAct's integrated agentic AI system (viGent) transforms construction safety through multilayered protection: computer vision detects accidents and unsafe behaviors (slips, falls, and personal protective equipment violations), while AI video analytics enable real-time monitoring via existing CCTV. The system's predictive capabilities analyze historical/real-time data to preempt risks: during emergencies, it autonomously maps evacuation routes and alerts personnel. For vehicle operations, viGent prevents collisions by dynamically adjusting forklift paths and logging near-miss data to optimize traffic flow. In confined spaces, it monitors environmental thresholds and worker duration, triggering automatic evacuations when hazards exceed limits. This centralized platform provides safety officers with comprehensive oversight while enabling immediate AI-driven interventions that surpass human response capabilities.[25]

and IoT integration, they dynamically analyze real-time data to predict hazards and enact proactive responses, like instantly alerting workers of slippery floors near machinery while rerouting foot traffic and notifying supervisors.[20] In the construction industry, for instance, agentic AI serves as a valuable tool for risk management by autonomously analyzing site hazards, optimizing workflows, and facilitating human-like decision making. This technology enhances safety protocols and operational efficiency, leading to more secure and productive construction environments.[21]

CHALLENGES AND WAYS FORWARD

The rise of AI agents offers significant automation benefits in HR, but it also introduces several challenges. The rapid proliferation of agentic AI without adequate governance can lead to inconsistent behaviors and decision making, complicating regulatory compliance and alignment with business objectives. For instance, the EU AI Act regulates all AI applications, including agentic AI, by risk level to ensure ethical, transparent, and accountable deployment. The EU AI Act mandates strict data protection for agentic AI, requiring data minimization and user control over personal data. High-risk applications must undergo bias assessments, rigorous testing, and external review before deployment.[22]

Agentic AI risks enabling adaptive "smart malware" that autonomously evades security defenses.[23] In HRM, agentic LLMs handling sensitive employee and candidate data present major security risks. Their extensive access to personal information and system controls makes them attractive targets for cyberattacks.[24]

Employee resistance to AI adoption persists, particularly in HR and administrative roles, where fears of job displacement remain prevalent.[27] Some employees remain hesitant to adopt AI agents, concerned about potential role displacement.[18] Despite proposals to integrate AI agents as "colleagues," HR professionals have largely rejected this approach. For example, Lattice's 2024 attempt to incorporate virtual workers into org charts sparked immediate backlash, forcing the company to abandon the initiative within days.[25]

Agentic AI's success hinges on high-quality, unbiased data; inaccuracies can lead to biased outcomes and erode trust. Ensuring data integrity is crucial to prevent reinforcing existing biases and to support effective AI adoption in HRM.[26]

To address these challenges, organizations must implement strong safeguards, such as clear governance frameworks, rigorous data controls, and enhanced cybersecurity measures. Human oversight remains essential to ensure AI aligns with ethical standards and business objectives.

Agentic AI is transforming HR by automating workflows, enhancing decision making, and improving employee experiences, from recruitment to safety compliance. While it boosts efficiency and productivity, challenges like data security, bias, and employee resistance require robust governance and human oversight. By balancing innovation with ethical safeguards, organizations can harness agentic AI's full potential to create strategic, adaptive, and human-centric HR systems for the future.

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NIR KSHETRI is a professor of management in the Bryan School of Business and Economics, University of North Carolina at Greensboro, Greensboro, NC 27412 USA. Contact him at nbkshetr@uncg.edu.